# Stablecoins as the Monetary Substrate for AI Agents: Architectures, Incentives, and Regulation in the Emerging Agentic Economy

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***Abstract*—Stablecoins have matured into a systemically relevant on-chain "cash layer" (a programmable, always-available settlement medium), with aggregate market capitalization exceeding $305.8B and dominant usage as settlement collateral across decentralized finance and cross-chain liquidity corridors (pathways enabling value transfer between separate blockchain networks). In parallel, recent large language models have achieved multi-step reasoning and autonomous tool-use capabilities, enabling a transition from passive information retrieval to autonomous economic agency. This evolution is operationalized through enterprise development environments (e.g., Claude Code, Cursor) and decentralized coordination platforms (e.g., Moltbook, OpenClaw), demonstrating a fundamental shift toward high-velocity machine-to-machine commerce. However, this autonomy amplifies systemic financial vulnerabilities: recent analysis reveals that 26.1% of agent skills contain at least one security vulnerability, spanning prompt injection, data exfiltration, and supply-chain risks.**

**This work examines the technical, economic, and regulatory dimensions of stablecoins as monetary infrastructure for AI agents. Our contributions include: (i) a five-layer architecture separating reasoning from spending authority through mandate-bounded authorization and independent verification; (ii) formal economic models incorporating budgeted autonomy and correlated risk; (iii) security guarantees with quantifiable $(\epsilon, \delta)$-bounds; (iv) quantitative risk assessment for stablecoin selection; (v) chain-aware regulatory compliance scoring methodology; and (vi) analysis of emerging web-native payment protocols including x402, with explicit treatment of their operational limits and deployment constraints.**

***Index Terms*—Stablecoins, AI Agents, Blockchain, Smart Contracts, Payment Systems, Multi-Agent Systems, Game Theory, Financial Regulation, Security**

## I. Introduction

Stablecoins are now sufficiently large to influence traditional financial markets through their reserve management. A Bank for International Settlements (BIS) working paper reports that as of March 2025, stablecoins' combined assets under management exceeded $200B, with approximately $40B in U.S. Treasury bills purchased during 2024—comparable to large U.S. money market funds [1]. Aggregate stablecoin market capitalization has reached $305.8B [33], with USDT representing approximately 60% market share. This scale has driven intensified regulatory oversight: the U.S. GENIUS Act establishes monthly reserve reporting, accounting-firm examination, and CEO/CFO certification requirements for payment stablecoin issuers, signaling a transition from experimental cryptocurrency to regulated payments infrastructure [6].

Concurrently, frontier large language models—including GPT-5.2 [16], Claude Opus 4.6, and Gemini 3.1 Pro [17]—have demonstrated multi-step reasoning, tool invocation, and autonomous planning capabilities sufficient to enable operational economic agency. This shift is increasingly operationalized through enterprise agentic development environments such as Claude Code [21] and Cursor [22], which allow models to manage complex repositories and execute system-level commands. The January 2026 release of Cursor 2.4 introduced a parallel subagent architecture enabling principal agents to delegate discrete subtasks to specialized subagents operating in independent context windows [23], mirroring broader trends in multi-agent orchestration research [8], [10], [11].

The rapid emergence of autonomous agent platforms with executable plugins and external tool access provides a real-world stress test of this autonomy. Recent large-scale security analyses reveal that 94.4% of state-of-the-art LLM agents are vulnerable to prompt injection, 83.3% to retrieval-based backdoors, and 100% to inter-agent trust exploits [12], while supply-chain vulnerabilities in agent tools (buffer overflow, SQL injection, XSS) create additional attack surfaces [13]. A complementary analysis of 31,132 agent skills found that 26.1% contain at least one vulnerability, spanning 14 distinct attack patterns across prompt injection, data exfiltration, privilege escalation, and supply-chain compromise [15]. The combination of access to private data, exposure to untrusted content, and capability for external financial action creates a critical vulnerability surface for agent-mediated payments.

The central question for agent ecosystems is not simply whether agents can transact, but rather: *under what architecture can agentic payment be simultaneously secure, economically rational, and compliance-conformant?* The same mechanisms enabling autonomous coordination create supply-chain attack surfaces for hijacking monetary transactions.

This work addresses this question through formal analysis and system design. We develop a five-layer architecture separating decision-making autonomy from spending authority through mandate-bounded authorization and independent verification, formalize economic models incorporating budgeted

autonomy and correlated risk structures, derive security guarantees with quantifiable $(\epsilon, \delta)$-bounds, provide quantitative frameworks for stablecoin selection under regulatory constraints, and analyze emerging web-native payment protocols such as x402 [31], including the operational conditions under which they are suitable for agent deployment [32].

## II. Related Work

### A. Stablecoin Design and Stability

Stablecoin scholarship has converged on a design taxonomy: fiat-reserve-backed, crypto-collateralized, algorithmic, and synthetic [2]. Algorithmic stablecoins (e.g., Terra/UST) maintain their peg through algorithmic supply adjustments and arbitrage incentives without external collateral, whereas synthetic stablecoins (e.g., Ethena/USDe) achieve stability via delta-neutral hedging strategies backed by derivative positions. Empirical literature emphasizes fragility in capital-efficient algorithmic designs. Analysis of the Iron Finance collapse demonstrates inherent fragility and run dynamics in algorithmic stablecoins, with transaction-level evidence paralleling Terra–Luna [3]. Formal analysis using computer-science abstractions argues that capital-efficient algorithmic decentralized stablecoins cannot be provably stable in general, highlighting "death spiral" vulnerabilities [4].

User perception studies indicate that "stable value" and "regulatory compliance" are perceived as key advantages, while concerns center on centralization (fiat-backed), complexity (algorithmic), and execution realism (crypto-backed) [5].

### B. AI Agents and Tool Use

The AI agent literature exhibits two main threads: (i) foundational architectures for tool-augmented agents [7], and (ii) multi-agent orchestration frameworks [8], [11]. Recent surveys characterize collaboration mechanisms based on actors, types (cooperation, competition, coopetition), structures (peer-to-peer, centralized, distributed), strategies (role-based, model-based), and coordination protocols [11]. Simulation work demonstrates that memory, reflection, and planning mechanisms produce emergent coordination from sparse initial goals [9]. These findings are significant for financial applications: once agents can transact, they can recruit other agents, outsource subtasks, and form market structures.

Security research identifies prompt injection and confused-deputy vulnerabilities as fundamental to tool-using agents, with amplified harm when tools include privileged actions [14], [19], [20]. Studies on cyber-physical agent systems emphasize context poisoning (malicious tool outputs influencing behavior), supply-chain exposure from unvetted connectors, and over-privileged credentials at integration boundaries [14]. Microsoft's LLMail-Inject challenge at IEEE SaTML explicitly targeted indirect prompt injection in agent-like environments [20].

### C. Regulatory Frameworks

Regulatory literature emphasizes macro-financial and AML/CFT dimensions. FATF treats stablecoins as potentially mass-adopted value transfer systems and emphasizes that existing standards apply [24], [25]. The Financial Stability Board's recommendations stress par redemption, governance accountability, risk management, and resolution planning [26]. These positions increasingly intersect with agentic economies where compliance cannot rely on human-in-the-loop verification.

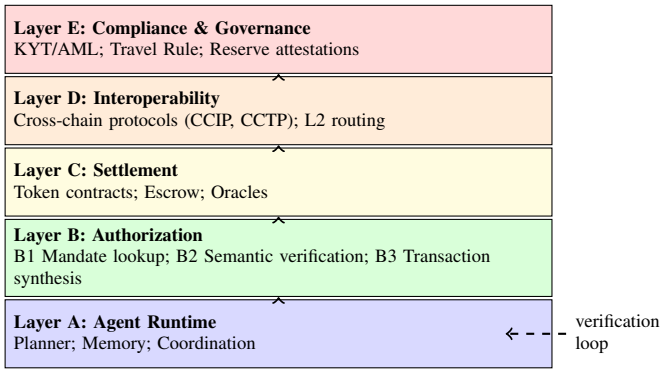



Fig. 1: Layered architecture for agent payment systems

## III. System Architecture

### A. Layered Design Principles

A secure agent–stablecoin architecture must explicitly separate *decision-making autonomy* from *spending authority*. This separation is compelled by observed attack patterns in autonomous agent systems: supply-chain vulnerabilities in executable plugins and unvetted tool connectors can trigger unauthorized financial transactions [12], [14].

Figure 1 illustrates the proposed five-layer architecture. Layer A (Agent Runtime) implements planning and tool selection but cannot directly control funds. Layer B (Authorization) enforces policy constraints, intent binding, and spending limits. Layer C (Settlement) handles on-chain execution. Layer D (Interoperability) manages cross-chain transfers. Layer E (Compliance) implements KYT/AML monitoring and regulatory reporting.

Crucially, Layer A is restricted to *intent proposal* rather than transaction construction. The authorization path therefore contains a deterministic mandate compiler and an independent verification loop before any payment payload is emitted. This addresses the confused-deputy problem: even if Layer A is compromised via prompt injection and proposes a malicious but syntactically valid transfer, Layer B binds the request to a pre-registered mandate (counterparty class, purpose code, maximum amount, expiry, and settlement domain) and rejects any payload whose semantics cannot be justified by verified external state (invoice, quote, subscription record, or escrow release condition).

### B. Formal Policy Language

We formalize payment policies to enable machine verification and composability.

**Definition III.1** (Payment Policy)**.** *A payment policy* $\Pi$ *is a tuple* $\langle \mathcal{C}, \mathcal{A}, \mathcal{L}, \mathcal{T} \rangle$ *where* $\mathcal{C}$ *is a set of conditions (amount bounds, recipient allowlists, temporal constraints),* $\mathcal{A}$ *is a set of actions (Allow, Deny, Delay, RequireApproval),* $\mathcal{L}$ *is a set of limits (per-transaction, daily, monthly), and* $\mathcal{T}$ *is a set of time-lock specifications.*

**Definition III.2** (Policy Satisfaction)**.** *A transaction* $\tau = (sender, recipient, amount, timestamp)$ *satisfies policy* $\Pi$ *relative to execution history* $\mathcal{H}$*, written* $(\tau, \mathcal{H}) \models \Pi$*, if:*

$$\begin{aligned} &\forall c \in \mathcal{C} : c(\tau) \\ &\quad \wedge \sum_{\tau' \in \mathcal{H}[window_\Pi(timestamp)]} \tau'.\text{amount} + \tau.\text{amount} \\ &\qquad \leq \mathcal{L}(period(timestamp)) \\ &\quad \wedge \ timelock_\Pi(timestamp) = expired \end{aligned} \tag{1}$$

Here, period : Timestamp $\to$ $\{\text{per-tx}, \text{daily}, \text{monthly}, \ldots\}$ maps a timestamp to its enclosing accumulation window, $\mathcal{H}[\text{window}_\Pi(timestamp)]$ denotes the set of previously executed transactions in that window, and $\mathcal{L}(\text{period}(timestamp))$ returns the maximum cumulative spend permitted within that window. This accumulated-spend form prevents a sequence of individually valid transactions from collectively bypassing the budget constraint.

**Theorem III.1** (Policy Composability)**.** *Given policies* $\Pi_1, \Pi_2$*, their conjunction* $\Pi_1 \wedge \Pi_2$ *forms a valid policy where* $(\tau, \mathcal{H}) \models (\Pi_1 \wedge \Pi_2) \iff (\tau, \mathcal{H}) \models \Pi_1 \wedge (\tau, \mathcal{H}) \models \Pi_2$.

*Proof.* Follows directly from conjunction of Boolean predicates over transaction attributes. □

**Definition III.3** (Mandate-Bound Authorization)**.** *A mandate* $M$ *is a deterministic tuple* $\langle purpose, \mathcal{R}, B, \chi, t_{\text{exp}} \rangle$ *consisting of an allowed purpose class, recipient set* $\mathcal{R}$*, spend bound* $B$*, admissible settlement domains* $\chi$*, and expiry* $t_{\text{exp}}$*. A payment request emitted by Layer A is authorized only if it can be compiled into a transaction* $\tau$ *such that (i)* $(\tau, \mathcal{H}) \models \Pi$*, (ii)* $\tau$ *is consistent with some active mandate* $M$*, and (iii) an independent verifier confirms that the referenced business object (invoice, quote, escrow condition, or subscription state) matches* $\tau$*.*

This refinement shifts the architecture away from direct LLM-to-payment integration. The model captures user or organizational intent and proposes an action, but a deterministic compiler and verifier synthesize the transfer only from structured mandate fields and verified external state. In effect, the payment rail follows a mandate-based architecture rather than a free-form generative one, aligning with emerging agent-payment designs based on cryptographically anchored intent verification [44].

### C. Cross-Chain Settlement Analysis

**Definition III.4** (Settlement Latency)**.** *Total settlement time for cross-chain transfer is:*

$$T_{\text{settle}} = T_{\text{conf}}^{\text{src}} + T_{\text{att}} + T_{\text{relay}} + T_{\text{mint}}^{\text{dst}} + \epsilon_{\text{net}} \tag{2}$$

*where* $T_{\text{conf}}^{\text{src}}$ *is source chain confirmation time,* $T_{\text{att}}$ *is attestation service processing,* $T_{\text{relay}}$ *is relayer submission time,* $T_{\text{mint}}^{\text{dst}}$ *is destination chain confirmation, and* $\epsilon_{\text{net}}$ *is network propagation delay. While the formula is general, each component is chain-specific: consensus mechanism, block time, and finality guarantees vary substantially across L1s and L2s.*

**Proposition III.2** (Settlement Bounds)**.** *We illustrate with Ethereum PoS as source chain, where the distinction between confirmation and finality is well-studied [39]:*

- ***Confirmed****: 2 blocks (≈26s) provides probabilistic safety under majority-honest validators. Recent research on fast confirmation rules demonstrates that vote-counting approaches can achieve confirmation in ≈69s with bounded adversarial assumptions [39].*
- ***Finalized****: 2 epochs (64 slots ≈ 768s ≈ 12.8 minutes) achieves economic finality via two-thirds validator supermajority.*

*For optimistic rollup destinations:*

$$T_{\text{settle}}^{\text{fast}} \approx 26s + T_{\text{att}} + T_{\text{relay}} + 2s \quad \textit{(confirmed)} \tag{3}$$

$$T_{\text{settle}}^{\text{sec}} \approx 768s + T_{\text{att}} + T_{\text{relay}} + T_{\text{fp}} + T_{\text{val}} \quad \textit{(finalized + proof/attestation delays)} \tag{4}$$

*Here* $T_{\text{fp}}$ *is the chain-specific fraud/fault-proof challenge window.* $T_{\text{val}}$ *captures any additional validity-proof or attestation latency imposed by the bridge or interoperability protocol. For major optimistic rollups,* $T_{\text{fp}}$ *is often on the order of 7 days [40], but we keep the term symbolic because the value is protocol-dependent and may change over time. More recent work on dynamic fraud-proof regimes reinforces that this window should be treated as a design parameter rather than a universal constant [41].*

Agents prioritizing speed may accept confirmation-based settlement for small transactions, while high-value settlements require finality-based guarantees to resist reorganization attacks.

### D. Cryptographic Custody

We employ $(m, n)$-threshold signature schemes where the private key is secret-shared into $n$ shares with any $m$ shares sufficient to reconstruct a signature. Let $a$ denote the number of automated participants (agent runtime, policy engine, and any additional programmatic signers). Shares are distributed as follows: $a$ shares across automated components (e.g., 1 for agent runtime, 1 for policy engine), 1 share for human override, and $n - a - 1$ shares held in Hardware Security Module (HSM) backup. The reconstruction threshold is set to $m = a + 1$, ensuring that the human override share is *mandatory* for high-risk authorization events such as key rotation, allowance replenishment, or transactions exceeding the active mandate ceiling. Routine low-risk payments that remain within pre-authorized mandates can instead be executed from a pre-funded allowance pool governed by the policy engine and automated signers, preserving bounded agentic spending

without requiring a human co-sign on every transfer. The HSM backup shares ($n - a - 1$) serve as recovery mechanism, but require human participation to reach the threshold, preventing autonomous recovery. Security property: compromise of $< m$ shares reveals no information about the private key under standard computational assumptions [29].

### E. Web-Native Micropayments

An emerging integration pattern enables "payments over HTTP" rather than payments embedded solely in decentralized applications. The x402 protocol [31] operationalizes HTTP 402 ("Payment Required") as a programmatic payment negotiation mechanism: if a request lacks payment, the server returns 402; the client then constructs payment programmatically and retries with a payment payload. This design explicitly targets APIs, websites, and autonomous agents with minimal integration overhead.

The x402 protocol has demonstrated real-world viability for high-frequency machine commerce: as of the January 2026 dashboard snapshot, self-reported network metrics indicate over 35 million transactions representing approximately $10 million in cumulative volume, primarily on Solana [32].[1] However, this should not be read as an unconditional endorsement for agent payments. The protocol introduces a two-phase request–payment–retry flow that increases end-to-end latency and creates an atomicity gap between service request and settlement confirmation. In addition, relay economics may require off-protocol compensation, early EVM realizations were constrained by token standards such as EIP-3009 and therefore excluded major stablecoins lacking compatible authorization flows, and naive implementations that rely on broad standing allowances create avoidable approval-risk exposure; this is the same general attack surface long associated with unlimited ERC-20 approvals to routers or facilitators [31].

The compliance-relevant feature is x402's anticipation of "facilitators" that verify and execute payments. Production facilitators can apply KYT/OFAC compliance checks and require API authentication, providing a natural integration point for regulatory compliance in agent deployments without modifying agent logic directly. Our revised position is therefore conditional: x402 is appropriate for low-value API commerce only when (i) the added latency is acceptable, (ii) the asset is natively supported, (iii) relay fees are explicitly priced, and (iv) approvals are exact-amount, short-lived, or otherwise cryptographically scoped rather than unlimited.

Early EVM implementations were effectively constrained by EIP-3009-style authorization flows, which excluded tokens such as USDT and DAI. More recent specifications add broader fallback paths, including Permit2-style authorization, substantially widening ERC-20 compatibility even though operational and approval-scope risks remain.

[1]While x402 currently operates primarily on Solana, the protocol specification supports multi-chain deployment and EVM-compatible implementations are under development. Reported metrics reflect the Solana deployment through January 2026.

### F. Multi-Agent Topology Constraints

Multi-agent orchestration should be treated as an architectural constraint rather than an incidental implementation detail. Recent empirical work shows that error amplification in multi-agent systems is strongly topology-dependent and can greatly exceed single-agent baselines when coordination is left unchecked [43]. If a workflow of depth $d$ has per-step verification-adjusted success probability $p_v$, end-to-end success scales as $p_v^d$, so even moderately reliable components degrade rapidly in long unchecked chains. For this reason, payment-capable systems should enforce bounded-depth execution graphs and require any delegated output that can influence authorization to be normalized into structured claims and re-checked by an independent verifier before it reaches Layer B.

## IV. Economic Modeling

### A. Time-Dependent Agent Optimization

An autonomous agent must decide which tasks to pursue over time to maximize profit while managing risk. The agent selects a time-dependent task portfolio $\mathcal{A}(t)$, where $t \in [0, T]$ denotes continuous time and $T$ is the planning horizon endpoint. At each instant $t$, the portfolio $\mathcal{A}(t)$ specifies which tasks the agent actively pursues. Each task $i$ yields revenue $r_i(t)$, incurs off-chain costs $c_i(t)$, and on-chain costs $f_i(t)$. We define the *net payoff* per task as:

$$X_i(t) = q_i(t)\, r_i(t) - c_i(t) - f_i(t) \tag{5}$$

where $q_i(t) \in [0, 1]$ is the success probability of task $i$ at time $t$. The agent's objective is to maximize the *expected present value* of the aggregate payoff stream over the horizon $[0, T]$, minus a risk penalty:

$$\max_{\mathcal{A}(t)} \mathbb{E}\left[\int_0^T e^{-\delta t} \sum_{i \in \mathcal{A}(t)} X_i(t)\, dt\right] - \mathcal{R}(\mathcal{A}) \tag{6}$$

Here, $e^{-\delta t}$ is the continuous-time discount factor (with rate $\delta > 0$), so the integral computes the present value of future net payoffs—analogous to discounted cash flow valuation of multiple investment streams over period $T$. The term $\mathcal{R}(\mathcal{A})$ captures aggregate risk (defined in Section IV-B).

Subject to:

$$\sum_{i \in \mathcal{A}(t)} f_i(t) \le B_{\text{gas}}(t) \qquad \forall t \in [0, T] \tag{7}$$

$$\sum_{i \in \mathcal{A}(t)} (c_i(t) + f_i(t)) \le W(t) \qquad \forall t \in [0, T] \tag{8}$$

$$q_i(t) \ge q_{\min} \qquad \forall i, t \tag{9}$$

$$|\mathcal{A}(t)| \le N_{\max} \qquad \forall t \tag{10}$$

where $B_{\text{gas}}(t)$ is the gas budget at time $t$, $W(t)$ is wallet balance, $q_{\min}$ is the minimum acceptable success probability, and $N_{\max}$ bounds concurrency.

### B. Correlated Risk Model

Not all risks are independent. Drawing from the threat taxonomy in Section V-A, an agent faces four primary loss sources per task: compromise of the agent itself, stablecoin depeg, cross-chain bridge failure, and regulatory penalty. These represent the dominant financial risk channels, though operational risks (e.g., oracle failure, gas price spikes) may also apply in specific deployments. Critically, tasks using the same stablecoin share systematic risk—if one task suffers a depeg loss, others using that coin likely will too.

**Definition IV.1** (Risk Functional)**.** *The risk functional $\mathcal{R}(\mathcal{A})$ decomposes as:*

$$\mathcal{R}(\mathcal{A}) = \underbrace{\sum_{i\in\mathcal{A}} \left(\lambda L_i^{\text{agt}} + p_i^{\text{dep}} L_i^{\text{dep}} + p_i^{\text{xc}} L_i^{\text{xc}}\right)}_{\textit{expected loss}} + \underbrace{\sum_{i\in\mathcal{A}} p_i^{\text{reg}} L_i^{\text{reg}}}_{\textit{regulatory loss}} + \rho(\mathcal{A}) \quad (11)$$

*where: $\lambda$ is the probability of agent compromise (from Section V); $L_i^{\text{agt}}$ is the expected monetary loss from agent compromise on task $i$; $p_i^{\text{dep}}$ and $L_i^{\text{dep}}$ are the depeg probability and loss (estimated from historical data in Table II); $p_i^{\text{xc}}$ and $L_i^{\text{xc}}$ are cross-chain bridge failure probability and loss; $p_i^{\text{reg}}$ and $L_i^{\text{reg}}$ are regulatory penalty probability and loss; and*

$$\rho(\mathcal{A}) \;=\; \gamma \sqrt{\sum_{\substack{i,j\in\mathcal{A}\\ i<j}} \text{Cov}(L_i, L_j)\cdot \mathbb{1}[\text{same-coin}(i,j)]} \quad (12)$$

*captures a reduced-form correlation penalty for systematic depeg contagion across tasks sharing the same stablecoin. The square root converts the covariance aggregate into monetary units, while $\gamma \geq 0$ is a dimensionless risk-aversion parameter and the restriction $i < j$ counts each pair exactly once.*

**Assumption IV.1** (Depeg Contagion)**.** *For tasks $i, j$ using the same stablecoin (algorithmic or synthetic):*

$$\Pr[L_j^{\text{dep}} > 0 \mid L_i^{\text{dep}} > 0] \geq \Pr[L_j^{\text{dep}} > 0] \quad (13)$$

*reflecting correlated failure modes: death-spiral dynamics for algorithmic designs and simultaneous hedging strategy failure for synthetic designs.*

### C. Budgeted Autonomy

A principal (human or organization) must decide how much spending authority to delegate to an agent. Too little budget creates operational friction (the agent must frequently request approval), while too large a budget increases exposure to loss. The optimization balances these two forces:

$$\min_{B\geq 0} \mathbb{E}[C^{\text{fr}}(B)] + \lambda_B\, B \quad (14)$$

Subject to:

$$\Pr[\text{total loss in } [0,T]] \leq \epsilon_{\text{risk}} \quad (15)$$

$$B \leq B_{\max}, \quad C^{\text{fr}}(B) \leq C_{\max} \quad (16)$$

where $C^{\text{fr}}(B)$ denotes friction cost, modeled as $\alpha B^{-\beta}$ with $\beta \in (0,1)$, and $\lambda_B > 0$ is the marginal cost of budget exposure (distinct from the agent compromise probability $\lambda$ in Definition IV.1).

**Lemma IV.1** (Optimal Budget)**.** *The optimal budget satisfies:*

$$B^* = \left(\frac{\alpha\beta}{\lambda_B}\right)^{\frac{1}{\beta+1}} \quad (17)$$

*when constraints are non-binding.*

*Proof.* First-order condition: $-\alpha\beta B^{-(\beta+1)} + \lambda_B = 0$ yields the result directly. □

### D. Multi-Agent Resource Allocation Game with Congestion Externalities

We model inter-agent competition as a multi-agent resource allocation game with congestion externalities rather than a strict Rosenthal congestion game. Each agent allocates effort across a finite task set, and payoffs deteriorate as aggregate load increases on shared payment, compute, or liquidity resources.

**Definition IV.2** (Agent Utility)**.** *Let agent $i$ choose a continuous allocation vector $x_i \in X_i := \{x_i \in [0,1]^M : \sum_{m=1}^M x_{im} \leq 1\}$ over $M$ task classes. Agent $i$'s utility in the $N$-player game (stated in static form for equilibrium analysis, abstracting from the time-dependent formulation in Section IV-A) is:*

$$U_i(x_i, x_{-i}) = \sum_{m=1}^{M} x_{im}\, v_{im}(S_m) - \mathcal{R}_i(x_i) - \kappa \sum_{m=1}^{M} x_{im} S_m \quad (18)$$

*where $S_m = \sum_{\ell=1}^N x_{\ell m}$ is aggregate load on task class $m$, $v_{im}(S_m)$ is continuous in $S_m$ and concave in $x_{im}$ for fixed $x_{-i}$, and $\kappa > 0$ captures congestion externalities.*

Here $x_{im}$ denotes the fraction of agent $i$'s compute and capital budget allocated to task class $m$, so the continuous strategy space is an operational abstraction of resource allocation rather than a claim that a payment task is literally executed fractionally.

**Theorem IV.2** (Equilibrium Existence)**.** *Under standard convexity and continuity conditions, a pure-strategy Nash equilibrium exists in the multi-agent economy.*

*Proof.* Each $X_i$ is a non-empty, compact, convex simplex subset of $\mathbb{R}^M$. By construction, $U_i(x_i, x_{-i})$ is continuous in $(x_i, x_{-i})$ and concave in $x_i$ for fixed $x_{-i}$ because $v_{im}$ is concave in own allocation and $\mathcal{R}_i$ is convex. Therefore the best-response correspondence $B_i(x_{-i}) =$

$\arg\max_{x_i \in X_i} U_i(x_i, x_{-i})$ is non-empty, convex-valued, and upper hemicontinuous. The product correspondence $B(x) = \prod_i B_i(x_{-i})$ on $X = \prod_i X_i$ satisfies Kakutani's conditions, so there exists $x^* \in X$ with $x^* \in B(x^*)$, i.e., a pure-strategy Nash equilibrium in continuous allocations [30]. □

### E. Liquidation Risk

For crypto-collateralized stablecoins (e.g., DAI, LUSD), stability depends on the value of locked collateral exceeding outstanding debt. We model the probability that collateral value drops below the minimum required ratio, which would trigger liquidation.

**Definition IV.3** (Collateralization Ratio)**.**

$$\mathit{CR}(t) = \frac{V_{\text{coll}}(t) \cdot P_{\text{coll}}(t)}{V_{\text{debt}}(t) \cdot P_{\text{stable}}(t)} \tag{19}$$

*where $V_{\text{coll}}(t)$ is the quantity of collateral tokens locked, $P_{\text{coll}}(t)$ is the market price of the collateral asset (e.g., ETH), $V_{\text{debt}}(t)$ is the outstanding stablecoin debt, and $P_{\text{stable}}(t)$ is the stablecoin price (nominally \$1). Let $\mathit{CR}_0 = \mathit{CR}(0)$ denote the initial collateralization ratio.*

Under geometric Brownian motion $dP_{\text{coll}} = \mu P_{\text{coll}} dt + \sigma P_{\text{coll}} dW$ (where $\mu$ is the drift rate, $\sigma$ is volatility, and $W$ is a standard Wiener process):

**Proposition IV.3** (Collateralization Risk at Maturity)**.** *The probability that collateralization falls below the minimum ratio by time $T$ is:*

$$\Pr[\mathit{CR}(T) < \mathit{MCR}] = \Phi\left(\frac{\ln(\mathit{MCR}/\mathit{CR}_0) - (\mu - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \tag{20}$$

*where $\Phi$ is the standard normal CDF and MCR is the minimum collateralization ratio. Under the endpoint-monitoring model used here—where collateralization is checked only at time $T$—this expression gives the exact probability of horizon shortfall, i.e., risk at maturity rather than first-passage liquidation risk under continuous monitoring.*

*Proof.* Under GBM, $\ln(\text{CR}(T)/\text{CR}_0) \sim \mathcal{N}\left((\mu - \frac{1}{2}\sigma^2)T, \sigma^2 T\right)$. Hence

$$\Pr[\text{CR}(T) < \text{MCR}] = \Pr\left[\ln\left(\frac{\text{CR}(T)}{\text{CR}_0}\right) < \ln\left(\frac{\text{MCR}}{\text{CR}_0}\right)\right],$$

and standardization yields the stated expression with the drift term entering with a minus sign. □

## V. Security Framework

### A. Threat Model

We consider five compromise modes based on recent taxonomies of agentic AI security [12], [13]: (1) prompt injection embedding hidden instructions, (2) tool API exploitation, (3) memory/context poisoning, (4) supply-chain compromise via dependencies, and (5) credential leakage.

The primary architectural risk is the confused-deputy problem: a compromised planner can generate a transaction request that is perfectly formatted yet semantically unauthorized [18]. Accordingly, our threat model assumes that Layer A outputs are adversarially influenceable and must never be treated as sufficient evidence of payment intent.

### B. Security Guarantees

We define a quantitative security notion for agent payment systems. Informally, $\epsilon$ bounds the probability of a large unauthorized transfer occurring, while $\delta$ bounds the expected monetary loss if a compromise does occur.

**Definition V.1** ($(\epsilon, \delta)$-Security)**.** *An agent is $(\epsilon, \delta)$-secure if:*

$$\Pr[\textit{unauthorized transfer} > B \textit{ in period } T] \leq \epsilon \tag{21}$$

$$\mathbb{E}[\textit{loss} \mid \textit{compromise}] \leq \delta \tag{22}$$

**Theorem V.1** (Layered Defense Upper Bound)**.** *For $k$ defense layers with per-layer failure probability $p$:*

$$\Pr[\textit{breach}] \leq p^k \tag{23}$$

*under independence.*

In practice, shared dependencies (foundational model weights, shared SDKs, single-point RPC providers) induce positive correlation across layers. We extend the bound using a common-shock model from reliability engineering: let $q$ denote the probability of a systemic event (e.g., zero-day in the foundational model, infrastructure provider compromise) that simultaneously bypasses all $k$ layers.

**Theorem V.2** (Correlated Defense Bound)**.** *Under the common-shock model with systemic failure probability $q$ and independent per-layer failure probability $p$:*

$$\Pr[\textit{breach}] = 1 - (1 - q)(1 - p^k) \tag{24}$$

*For small $p$ and $q$, this simplifies to $\Pr[\textit{breach}] \approx q + p^k$, showing that systemic risk $q$ dominates the independent layered bound $p^k$ when $q \gg p^k$.*

*Proof.* A breach occurs either via the common shock (probability $q$) or via independent failure of all $k$ layers in the absence of a common shock (probability $(1 - q) \cdot p^k$). By inclusion: $\Pr[\text{breach}] = q + (1 - q)p^k = 1 - (1 - q)(1 - p^k)$. □

**Corollary V.3.** *With $p = 0.1$, $k = 5$ layers, and $q = 0.01$ (estimated systemic risk from shared infrastructure), $\Pr[\textit{breach}] \approx 0.01 + 10^{-5} \approx 0.01$. The systemic component dominates by three orders of magnitude, demonstrating that adding layers yields diminishing returns without addressing common-mode failures. This motivates architectural diversity: layers should minimize shared dependencies (e.g., distinct signing backends, heterogeneous runtime environments) to reduce $q$.*

### C. Formal Verification Properties

**Definition V.2** (Safety)**.** *The system is safe if:*

$$\forall \tau, \mathcal{H} : (\tau \textit{ executed with history } \mathcal{H}) \implies ((\tau, \mathcal{H}) \models \Pi \wedge \textit{balance}(\textit{sender}) \geq \tau.\textit{amount}) \tag{25}$$

**Definition V.3** (Liveness)**.** *The system is live if:*

$$\forall \tau, \mathcal{H} : ((\tau, \mathcal{H}) \models \Pi) \implies \Diamond(\tau \text{ executed within timeout}) \tag{26}$$

### D. Intent Preservation and Verification Loop

Layer isolation is meaningful only if authorization verifies *why* a payment is being made, not merely whether it satisfies scalar bounds. We therefore refine Layer B into three deterministic stages: (i) mandate lookup, which binds the request to a pre-authorized purpose and counterparty class; (ii) semantic verification, which checks the referenced invoice, quote, escrow predicate, or subscription record against the mandate; and (iii) transaction synthesis, which emits the settlement payload only from verified fields. This prevents Layer A from directly supplying arbitrary calldata or recipient parameters.

This also addresses the operational failures of direct LLM-to-payment pipelines. Rather than allowing the model to generate transaction-ready instructions, the model is limited to intent capture and structured claim extraction, while deterministic components perform authorization and settlement construction.[2] The result is a mandate-based control plane in which probabilistic model outputs cannot by themselves trigger payment, following the principle of cryptographically anchored intent verification [44].

## VI. Regulatory Compliance

### A. Compliance Scoring

**Definition VI.1** (Compliance Score)**.** *For stablecoin $s$ on chain $c$ in jurisdiction $j$:*

$$C_{\text{score}}(s, c, j) = \phi_{\text{chain}}(s, c, j) \sum_{i=1}^{5} w_i \cdot I_i(s, j) \tag{27}$$

*where issuer-level indicators $I_1, \ldots, I_5 \in [0, 1]$ represent: $I_1$ = licensed status, $I_2$ = third-party audit frequency, $I_3$ = real-time attestation availability, $I_4$ = KYT/AML integration, $I_5$ = travel rule compliance. The chain factor $\phi_{\text{chain}}(s, c, j) \in [0, 1]$ discounts the issuer baseline for deployment-specific frictions including native-vs-bridged issuance, chain-specific blacklist/enforcement support, validator/sequencer concentration, bridge dependence, and local travel-rule tooling. Weights satisfy $\sum_{i=1}^{5} w_i = 1$; equal weighting yields $w_i = 0.2$ for all $i$. For binary indicators (e.g., licensed vs unlicensed), values are $\{0, 1\}$. For continuous indicators (e.g., audit frequency), values are scaled to $[0, 1]$ by coverage or frequency relative to best-in-class standards.*

Table I presents *issuer-level baseline* scores for major stablecoins in U.S. jurisdiction with equal weights, derived from publicly verifiable regulatory filings, attestation reports, and compliance infrastructure as of Q1 2026.[3]

TABLE I: Issuer-Level Baseline Compliance Scores (U.S. Jurisdiction)

| Coin | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | Score |
|---|---|---|---|---|---|---|
| USDC | 1.0 | 1.0 | 0.5 | 1.0 | 1.0 | 0.90 |
| PYUSD | 1.0 | 1.0 | 0.0 | 1.0 | 1.0 | 0.80 |
| FRNT | 1.0 | 1.0 | 0.0 | 0.6 | 0.6 | 0.64 |
| USDT | 0.4 | 0.5 | 0.0 | 1.0 | 0.3 | 0.44 |
| DAI | 0.0 | 0.8 | 1.0 | 0.0 | 0.0 | 0.36 |
| USDe | 0.0 | 0.4 | 1.0 | 0.0 | 0.0 | 0.28 |

To illustrate the chain-sensitive adjustment, Ethereum-native USDC may warrant $\phi_{\text{chain}} \approx 1$, whereas bridged representations or chains with weaker issuer control, fragmented compliance tooling, or added bridge assumptions should receive a downward adjustment. Similarly, USDT on Ethereum and USDT on Tron should not be treated as compliance-equivalent deployments even when the issuer-level baseline is the same: BlockSec's 2025 analysis reports that the large majority of USDT blacklist actions occurred on Tron, indicating materially different operational compliance environments across deployment chains [45].

### B. Regulatory Landscape

The U.S. GENIUS Act mandates monthly reserve reporting with CEO/CFO certification under criminal penalty [6]. EU MiCA classifies asset-referenced and e-money tokens under Titles III and IV [28]. FATF emphasizes that existing AML/CFT standards apply to stablecoin arrangements [24], [25]. However, implementation remains uneven: 75% of jurisdictions are only partially compliant, with over half lacking travel rule implementation [27].

## VII. Comparative Analysis

Table II presents risk and performance metrics for major stablecoins. Depeg probabilities ($p^{\text{dep}}$) are derived from historical price data: we define a depeg event as a deviation >2% from

[2]While empirical benchmarks of LLM-to-payment failure rates under various architectures are an active area of study, to our knowledge no peer-reviewed measurement of the specific rates quoted in the review letter has been published as of Q1 2026. Our architectural mitigation is motivated by the theoretical risk profile—probabilistic outputs should not directly formulate financial transactions—rather than by any single benchmark.

[3]Scoring methodology: $I_1$: 1.0 = FinCEN MSB + NYDFS BitLicense or state-issued; 0.4 = MSB registration only; 0.0 = no U.S. license. $I_2$: 1.0 = monthly third-party attestation (e.g., Deloitte, KPMG); 0.8 = real-time on-chain verifiability; 0.5 = quarterly attestation; 0.4 = weekly proof-of-reserves. $I_3$: 1.0 = on-chain proof-of-reserves; 0.5 = partial (bridged tokens only); 0.0 = off-chain only. $I_4$: 1.0 = dedicated KYT provider (e.g., Chainalysis) or NYDFS-mandated program; 0.6 = via distribution partners only; 0.0 = none at protocol level. $I_5$: 1.0 = issuer-level Travel Rule compliance; 0.6 = via regulated distribution partners; 0.3 = in progress (e.g., Notabene trial); 0.0 = no implementation. $\phi_{\text{chain}}$: 1.0 = native issuance on a chain with first-party compliance tooling; values below 1.0 apply to bridged or operationally weaker deployments. Sources: Circle Transparency [34], Paxos PYUSD Reports [35], Wyoming Stable Token Commission [36], Tether Attestation Reports [37], Chainalysis KYT [38].

**Algorithm 1** Stablecoin Selection

**Require:** Risk tolerance $\epsilon$, jurisdiction $j$, use case $u$
**Ensure:** Selected stablecoin-chain pair $(s^*, c^*)$

$\mathcal{S} \leftarrow \{\text{available stablecoins}\}$
$\mathcal{S} \leftarrow \{(s,c) : C_{\text{score}}(s,c,j) \geq \theta_{\text{comp}}\}$
$\mathcal{S} \leftarrow \{(s,c) \in \mathcal{S} : p_s^{\text{dep}} + p_s^{\text{liq}} \leq \epsilon\}$
**if** $u =$ micropayments **then**
  $\mathcal{S} \leftarrow \{(s,c) \in \mathcal{S} : \text{cost}_{s,c} < 0.1\}$
**else if** $u =$ settlements **then**
  $\mathcal{S} \leftarrow \{(s,c) \in \mathcal{S} : C_{\text{score}}(s,c,j) > 0.9\}$
**end if**
$(s^*, c^*) \leftarrow \arg\max_{(s,c)\in\mathcal{S}} [\text{utility}(s,c) - \text{risk}(s,c)]$
**return** $(s^*, c^*)$

the \$1.00 peg sustained for >1 hour, and compute annualized event frequency over each token's operational history.[4]

### A. Selection Algorithm

Algorithm 1 formalizes stablecoin selection for agent deployments.

## VIII. Discussion

### A. Implementation Considerations

Deployment should proceed incrementally: (1) implement Layer B wallet with EIP-4337 smart accounts, deterministic mandate compiler, and basic policy engine (months 1–3); (2) add independent semantic verification, cross-chain support via CCTP, and threshold signatures (months 4–6); (3) integrate multi-stablecoin portfolio management with dynamic selection and bounded-depth multi-agent orchestration (months 7–12).

### B. Open Challenges

Key research directions include: formal verification of policy safety and liveness using tools such as TLA+ or Coq; adaptive risk models incorporating online learning for real-time parameter updates; privacy-preserving compliance mechanisms using ZK-SNARKs; analysis of emergent market structures in agent-dominated economies; standardized machine-readable compliance requirements; defense-in-depth architectures resilient to combined prompt injection and financial tool access; probabilistic bounds on cross-chain bridge failures; and optimal economic mechanism design for agent transaction fees, insurance, and dispute resolution.

[4]Depeg event data: USDC depegged to \$0.87 during the SVB crisis (March 2023, 48h duration) [46]; USDT hit \$0.9455 during Terra contagion (May 2022, 24h) [47]; DAI traded above \$1.10 during Black Thursday (March 2020) with \$8.32M in collateral liquidated for 0 DAI [48]; PYUSD and FRNT have recorded no depeg events >2% since launch. USDe experienced a Binance-specific oracle anomaly (October 2025) reaching \$0.65 on Binance, though broader market deviation was <0.3%. The $p^{\text{dep}}$ column reports annualized event frequency expressed in percent per year. The $p^{\text{liq}}$ column reports illustrative one-year endpoint shortfall probabilities from Eq. (20), using representative parameter sets rather than issuer-disclosed risk models; for crypto-collateralized designs we use horizon $T = 1$ year and stylized assumptions on $\mu$, $\sigma$, $\text{CR}_0$, and MCR chosen to reflect conservative stressed conditions rather than calibrated production risk engines. Transaction costs reflect Ethereum L1 (\$0.20–2.00), L2s (\$0.005–0.10), and Solana (\$0.0001–0.001) as of Q1 2026. Settlement times are blockchain-specific, not stablecoin-specific.

### C. Limitations

Our analysis relies on several simplifying assumptions: (1) the common-shock security model (Section V.B) captures systemic correlation via a single parameter $q$, but real-world failure dependencies may exhibit more complex multivariate structure; (2) static risk parameters (market conditions evolve); (3) perfect policy enforcement and verifier correctness (implementation bugs create gaps); (4) rational agent behavior (real agents may exhibit bounded rationality or be adversarially controlled); (5) current regulatory frameworks (rules are rapidly evolving); (6) chain adjustment factors $\phi_{\text{chain}}$ remain partially judgment-based until richer empirical datasets are available; and (7) the architecture awaits adversarial stress-testing and production-scale load evaluation. Future work should include adversarial testing of prompt injection scenarios, policy bypass attempts, verifier disagreement handling, and real-world latency/cost overhead measurement under production load.

## IX. Conclusion

This work presents a formal framework for integrating stablecoins into autonomous AI agent systems. We contribute: (1) a layered architecture with mandate-bound authorization, semantic verification, and composability guarantees, (2) economic models incorporating temporal dynamics, correlation penalties, and game-theoretic interactions, (3) quantifiable security properties with ($\epsilon$, $\delta$)-bounds, (4) chain-aware regulatory compliance scoring methodology, and (5) quantitative risk assessment with decision algorithms.

The fundamental tension is clear: autonomy without containment creates scalable financial vulnerability. Our revised framework responds by separating intent capture from transaction construction, inserting an independent verification loop before authorization, and treating stablecoin selection as both a jurisdictional and chain-specific control problem. As stablecoins scale toward \$500B+ and AI agents proliferate, these integration mechanisms will become critical infrastructure for secure, compliant, and economically rational agent-mediated payments.

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TABLE II: Stablecoin Risk and Performance Metrics

| **Stablecoin** | **Stability Model** | $p^{\mathrm{dep}}$ (%/yr) | $p^{\mathbf{liq}}$ (%/yr) | $C_{\mathrm{score}}$ (0-1) | **Cost** (USD) | **Settle** (sec) | **Agent Ready** | **Notable Depeg Event** |
|---|---|---|---|---|---|---|---|---|
| USDC | Fiat | 0.25 | — | 0.90 | 0.01–2 | 0.4–768 | High | SVB $0.87 (Mar'23) |
| PYUSD | Fiat | <0.1 | — | 0.80 | 0.01–2 | 0.4–768 | High | None recorded |
| FRNT | Fiat (state) | <0.1 | — | 0.64 | 0.01–1 | 0.4–400 | High | None (102% coll.) |
| USDT | Fiat | 0.20 | — | 0.44 | 0.01–2 | 0.4–768 | Med-High | Terra $0.95 (May'22) |
| DAI | Crypto-coll. | 0.50 | 1.5 | 0.36 | 0.20–5 | 26–768 | Medium | BT $1.10+ (Mar'20) |
| USDe | Synthetic | 1.0 | 5.0 | 0.28 | 0.20–5 | 26–768 | Low | Oracle $0.65 (Oct'25) |


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